BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Binary and Denary

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Recall from earlier Computing

Answer all four questions in BLACK PEN before reading the explanations.

1. Which two digits are used in binary?

2. How many bits make one byte? Circle one answer: 4 / 8 / 16 / 32

3. Calculate 2 to the power of 4. Show your working.

4. Four ON/OFF switches: how many combinations are possible? Explain.

Today's objectives

Use 8-bit place values; convert binary to denary and back; explain why unsigned eight-bit numbers range from 0 to 255.

Key vocabulary - read, cover, recall

binary Base 2: uses 0 and 1	**denary** Base 10: uses digits 0-9
bit One binary digit	**byte** Eight bits
place value Value from digit position	**most significant bit** Highest-value bit on the left

Write: In your own words, explain what a byte is.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn & follow

Binary place values

Computers represent data using two states: 0 and 1. In binary, each column has double the value of the column on its right.

Eight-bit place values

128	64	32	16	8	4	2	1
0	0	0	0	0	0	0	0

Eight bits give 256 patterns (2 to the power of 8). The unsigned range is 0-255 because counting starts at zero.

I Do - Binary to denary

Convert 11001010 into denary. Add the place values where the bit is 1.

Worked example

128	64	32	16	8	4	2	1
1	1	0	0	1	0	1	0

128 + 64 + 8 + 2 = 202

I Do - Denary to binary

Convert denary 74 into eight-bit binary. Select 64 + 8 + 2.

Worked example

128	64	32	16	8	4	2	1
0	1	0	0	1	0	1	0

74 = 01001010. Always include all eight bits, including leading zeros.

We Do - Guided practice

Convert 178 to unsigned eight-bit binary, starting with 128.

Complete the eight-bit chart

128	64	32	16	8	4	2	1
_____	_____	_____	_____	_____	_____	_____	_____

178 = 128 + _____ + _____ + _____

Eight-bit answer:

Check: Does your answer use exactly eight digits?

Guide check: 47 = 32 + 8 + 4 + 2 + 1, so 47 = 00101111 in binary.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think & write

Code First - Predict > Write > Test > Improve

Start on paper. A print(...) instruction displays the calculated result.

Predict: What would print(8 + 2 + 1) display?

Write: Write a print(...) instruction to display denary 10 (binary 1010).

Test and improve: Change the code below so it displays denary 9 (binary 1001).

```
value = 8 + 2 + 1
print(value)
```

You Do - Independent practice

1. Convert 01101001 to denary. Show your addition.

2. Convert 191 to unsigned eight-bit binary. Show the place values.

3. Someone claims 154 = 10011001. Convert their answer to denary, explain their mistake and write the correct eight-bit binary for 154.

Thinking: Create an eight-bit number between 150 and 180 (excluding both limits) with exactly four 1s. Explain your place values.

Exit Ticket - True or False?

1. The largest unsigned eight-bit number is 256. ☐ T ☐ F

2. A 1 bit contributes its place value to the total. ☐ T ☐ F

3. print(8 + 2) would display 10. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Check pages 1-3. Tick correct answers. For an error, read the model answer and complete 'Try the question again' at the bottom.

Page 1 - Do Now

Q1	0 and 1	☐
Q2	8 bits in one byte	☐
Q3	16 (2 x 2 x 2 x 2)	☐
Q4	16 combinations: 2 x 2 x 2 x 2	☐

Page 1 - Vocabulary

Binary	Base-2 number system: digits 0 and 1	☐
Denary	Base-10 system: digits 0-9	☐
Bit	One binary digit	☐
Byte	Eight bits	☐
Place value	A digit's value depends on its position	☐
MSB	Highest-value bit on the left	☐

Page 2 - We Do

178	128 + 32 + 16 + 2 = 178	☐
Answer	10110010	☐

Page 3 - Code First

Predict	11	☐
Write	print(8 + 2) displays 10	☐
Improve	value = 8 + 1; print(value) displays 9	☐

Page 3 - You Do / Exit Ticket

Q1	01101001 = 64 + 32 + 8 + 1 = 105	☐
Q2	191 = 128 + 32 + 16 + 8 + 4 + 2 + 1 = 10111111	☐
Q3	10011001 = 153; 154 is 10011010	☐
Thinking	Example: 10101010 = 170, with exactly four 1s	☐
Exit	False; True; True	☐

Try the question again

Choose one answer to improve. All correct? Choose the hardest.

Copy the whole question:

Write a new answer in GREEN PEN. Show working if needed: